

Airline Delay Analysis – Business Insights & Findings

Dataset Overview

This analysis uses the Airline Delay Cause dataset covering U.S. airline operations from 2003 to 2022.

Dataset Summary

- 318,017 records
- 33 airlines
- 420 airports
- 20 years of operational data

The dataset includes flight volume, delayed flights, cancellations, diversions, and delay causes such as carrier, weather, NAS, security, and late aircraft delays.

Key Preliminary Findings

Initial exploration revealed several important patterns:

- Large airlines accumulated the highest total delay minutes due to operating significantly more flights.
 - Average delay per flight provided a more meaningful measure of operational efficiency than total delay minutes alone.
 - Delay performance varied considerably across years, airlines, airports, and seasons.
 - The year 2020 represented a major outlier because of the significant reduction in air travel during the COVID-19 pandemic.
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Question 1: What Causes Most Airline Delays?

Findings

Late aircraft delays were the largest contributor to overall delay minutes across the industry, followed by carrier delays and NAS (National Airspace System) delays.

Delay Cause Breakdown:

- Late Aircraft Delay: 37.9%
- Carrier Delay: 30.6%
- NAS Delay: 26.1%
- Weather Delay: 5.2%
- Security Delay: 0.2%

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Nearly 95% of all delay minutes were driven by late aircraft, carrier-related, and NAS delays. This suggests that operational efficiency and airspace congestion have a much greater impact on airline performance than weather or security-related disruptions.

Question 2: Which Airlines Experience the Highest Delay Burden?

Findings

The highest delay rates were observed among:

1. Atlantic Southeast Airlines – 24.47%
2. JetBlue Airways – 23.68%
3. Allegiant Air – 22.65%
4. Comair Inc. – 22.32%
5. Frontier Airlines – 22.20%

Among larger carriers, Delta Air Lines and Alaska Airlines recorded comparatively lower delay rates, while Hawaiian Airlines demonstrated the strongest performance overall.

Business Insight

Delay rates varied significantly across airlines, indicating substantial differences in operational efficiency, scheduling effectiveness, and network management practices.

Question 3: Which Airports Act as Operational Bottlenecks?

Findings

A comparison of airport traffic volume and average delay minutes revealed that several major hub airports consistently experienced the highest delay burdens.

Top-performing bottleneck airports included:

- Newark Liberty International Airport
- LaGuardia Airport
- San Francisco International Airport
- Chicago O'Hare International Airport
- John F. Kennedy International Airport

The scatter plot analysis showed that airports with both high traffic volumes and elevated average delays represent critical pressure points within the airline network.

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Delays occurring at major hub airports can affect thousands of downstream flights. Improving operational efficiency at these locations would likely generate benefits across the broader transportation network.

Question 4: How Do Delays Change Throughout the Year?

Findings

Strong seasonality was observed across the dataset.

Months with the highest delay levels:

- June
- July

- August
- December

Months with the lowest delay levels:

- September
- October
- November

December recorded the highest overall delay rate, while June produced the highest average delay minutes per flight.

Business Insight

Peak travel seasons place additional strain on airport and airline operations. Airlines can reduce disruption by increasing staffing, improving scheduling flexibility, and enhancing operational planning before these high-demand periods.

Question 5: How Have Airline Delays Changed Over Time?

Findings

Delay performance changed considerably between 2003 and 2022.

Key trends included:

- Rising delay levels during the mid-2000s.
- More stable performance throughout much of the 2010s.
- A sharp decline in delays during 2020 due to reduced flight activity.
- A noticeable rebound in delays during 2021 and 2022 as travel demand recovered.

Business Insight

Although delay levels fluctuate over time, long-term results suggest that capacity constraints and operational challenges continue to affect airline performance. Future improvements will depend on stronger network planning, operational resilience, and congestion management strategies.

Data Quality & Cleaning Summary

Several data validation checks were performed before analysis:

- Checked for missing values across key fields.
- Investigated records containing missing carrier information.
- Identified four records with missing carrier data, all occurring in August 2008.
- Verified that no duplicate airline-airport-month records existed.
- Confirmed valid month ranges (1–12) and year ranges (2003–2022).

A cleaned analysis table was created while preserving the original raw dataset.

Conclusion

The analysis shows that airline delays are driven primarily by operational and system-level factors rather than weather or security disruptions. Late aircraft delays, carrier-related delays, and NAS delays account for the vast majority of delay minutes across the industry.

Major hub airports experience the greatest delay burden, and delays follow clear seasonal patterns, particularly during summer and holiday travel periods. While performance improved temporarily during the pandemic, delay levels increased again as passenger demand recovered.

These findings suggest that airlines and airport operators would benefit most from investments in operational efficiency, aircraft turnaround management, congestion reduction, and proactive planning during peak travel seasons.